PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

# Development of a Smart Classroom and Timetable Scheduler System

**AN INTERNSHIP REPORT**

***Submitted by***


## KRISHNA B – 20221ISE0090


*Under the guidance of,*

## Mr.Libin Jacob P

## BACHELOR OF TECHNOLOGY

**IN**

## INFORMATION SCIENCE AND ENGINEERING

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

**PRESIDENCY UNIVERSITY**

**BENGALURU**

**APRIL 2026**

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# PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

## BONAFIDE CERTIFICATE

Certified that this report "Development of a Smart Classroom and Timetable Scheduler System" is a bonafide work of Krishna B (20221ISE0090), who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE AND ENGINEERING during 2025-2026.

**Mr. Libin Jacob P**
Project Guide
PSCS
Presidency University

**Mr. Sreehari T M**
Program Internship Coordinator
PSCS
Presidency University

**Dr. Md Ziaur Rahaman**
School Internship Coordinator
PSCS
Presidency University

**Dr. Gopal Krishna Shyam**
Head of the Department,
PSCS
Presidency University

**Dr. Duraipandian N**
Dean
PSCS / PSIS
Presidency University

**Name and Signature of the Examiners**

| Sl. No | Name | Signature | Date |
|---|---|---|---|
| 1 | | | |
| 2 | | | |

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# DECLARATION

I am a student of final year B.Tech in INFORMATION SCIENCE AND ENGINEERING, at Presidency University, Bengaluru, named Krishna B, hereby declare that the internship work titled "Development of a Smart Classroom and Timetable Management System" has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in INFORMATION SCIENCE AND ENGINEERING during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Name of the Student: Krishna B USN: 20221ISE0090 Signature: _____

PLACE: BENGALURU

DATE: 23- April 2026

# INTERNSHIP COMPLETION CERTIFICATE

Robowaves

**CIN: U72200KA2007PTC044701** **Date: 08/04/2026**

**TO WHOM SO EVER IT MAY CONCERN**

This is to certify that **Mr/Ms. Krishna B** bearing **USN – "**20221ISE0090**"** a student of **"Bachelor of Technology"** from **Presidency University, Itgalpura, Rajanukunte, Yelahanka.** had worked as intern and completed his/her internship in our company and worked on **"Web Technology"** from **01/01/2026 to 31/03/2026**. Successfully completed 3 months of internship under the guidance of **Mr. Pradeep (Robowaves (A Unit of Test Yantra Software Solutions (India) Pvt. Ltd.)**

During this period of this internship program with us we found he/she is sincere and hardworking.

We wish success for all his/her future endeavours.

**Yours Sincerely,**

**For Robowaves**
**(A Unit of Test Yantra Software Solutions (India) Pvt. Ltd.)**
**"System Generated letter no need of Signature"**

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Krishna B

# ABSTRACT

The increasing adoption of digital technologies in educational institutions has created a growing need for efficient and centralized academic management systems. Traditional methods of managing attendance, timetable scheduling, assignment handling, and academic communication are often time-consuming, error-prone, and difficult to manage efficiently. Existing academic management platforms provide limited integration of classroom management functionalities, leading to challenges in accessibility, communication, and academic record maintenance. This internship report presents the development of the "Smart Classroom and Timetable Scheduler (SCTS)," a web-based academic management system designed to address these challenges.

The project involved designing and implementing a centralized platform that integrates attendance management, timetable scheduling, assignment management, academic notifications, and role-based access control for Students, Faculty, and Admin users. The system was developed using the Django framework with Python for backend development and HTML, CSS, JavaScript, and Bootstrap for frontend design. The methodology focused on creating a secure, user-friendly, and responsive platform that simplifies academic operations and improves communication within educational institutions.

The results of this project include a functional web-based academic management system capable of managing attendance records, timetable schedules, assignment submissions, and academic notifications efficiently. The developed platform successfully demonstrates how digital technologies can improve academic management, reduce manual administrative work, and enhance accessibility to academic information. The project also supports Sustainable Development Goals such as SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure) by promoting technology-driven and efficient educational management practices.

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# ABBREVIATIONS

| Abbreviation | Full Form |
|---|---|
| API | Application Programming Interface |
| SCTS | Smart Classroom and Timetable Scheduler |
| CRUD | Create, Read, Update, Delete |
| CSS | Cascading Style Sheets |
| DBMS | Data Base Management System |
| HTML | Hypertext Markup Language |
| HTTP | Hypertext Transfer Protocol |
| Django ORM | Django Object Relational Mapper |
| DRF | Django Representational State Transfer Framework |
| JSON | JavaScript Object Notation |
| SQL | Structured Query Language |
| MVC | Model View Controller |
| Wi-Fi | Wireless Fidelity |
| REST | Representational State Transfer |
| SDG | Sustainable Development Goal |
| SSID | Service Set Identifier |
| POST | Protocol for HTTP Submission Method |
| UI/UX | User Interface / User Experience |
| URL | Uniform Resource Locator |
| VPN | Virtual Private Network |

# Chapter 1

# INTRODUCTION

## 1.1 Background

The increasing adoption of digital learning environments and smart educational technologies has encouraged institutions worldwide to modernize their academic management systems. Many educational institutions still rely on conventional methods for managing attendance, classroom schedules, assignment submissions, and academic communication. These outdated systems often lead to administrative delays, inaccurate record maintenance, lack of transparency, and increased workload for faculty and administrative staff.

The **Smart Classroom and Timetable Scheduler (SCTS)** was developed to address these limitations by creating a centralized academic management platform that combines multiple classroom management functionalities into a single system. The project aims to simplify daily academic operations while improving accessibility, communication, and operational efficiency within educational institutions.

The platform is designed with scalability, security, and usability in mind. By utilizing Django's secure authentication system and database management capabilities, the project ensures reliable data handling and role-based access control. Additionally, the system supports mobile accessibility, allowing users to access academic information anytime and from any device connected to the institutional network.

## 1.2 Statistics of Internship

**Organization:** RoboWave(A Unit of Test Yantra Software Solutions)

**Role:** Full Stack Web Developer Intern

**Reporting Manager:** Govind Raj

**Internship Duration:** 01-01-2026 to 31-03-2026

**Total Hours Completed:** 539 Hours

**Team Structure:** Small, agile startup team

**Primary Contributions:**

- Implementing digital attendance management features.
- Developing timetable scheduling and classroom management modules.
- Designing responsive web interfaces for students, faculty, and administrators.
- Testing and improving system performance and usability.

## 1.3 Prior Existing Technologies

The primary existing technologies in this domain are Learning Management Systems and academic management platforms:

**Google Classroom:** Used for assignment sharing, classroom communication, and online learning.It does not provide integrated timetable scheduling and advanced classroom management features.

**Model:** An open-source learning management system used for course management and online education. It requires complex setup and maintenance for institutions.

**ERP-Based Academic Systems:** Used for attendance tracking, timetable management, and student record handling. These systems are often costly and difficult to customize.

**Key Gaps Identified in Existing Technologies**

- Lack of integrated attendance and timetable management in a single platform.
- Limited real-time communication features.
- Complex user interfaces for students and faculty.
- High maintenance and deployment costs.

## 1.4 Proposed Approach

The proposed approach for the Smart Classroom and Timetable Scheduler is to develop a centralized academic management system that improves classroom operations through automation and digital management. The system focuses on simplifying attendance tracking, timetable scheduling, assignment management, and academic communication within a single platform.

The platform is developed using the Django framework with Python for backend development and HTML, CSS, JavaScript, and Bootstrap for the frontend interface. The system provides role-based access for Students, Faculty, and Admin users to ensure secure and organized management of academic activities.

## 1.5 Objectives

The primary objectives of the internship work were:

- To develop a Smart Classroom and Timetable Scheduler for academic management.
- To implement digital attendance tracking for students and faculty.
- To provide automated timetable scheduling and classroom management features.
- To enable assignment upload, submission, and academic notifications through a centralized platform.
- To implement secure role-based access for Students, Faculty, and Admin users.
- To create a responsive and user-friendly web application for academic activities..

## 1.6 SDGs

The development of the Amairo platform directly supports two United Nations Sustainable Development Goals (SDGs) [6]:

**SDG 9 – Industry, Innovation, and Infrastructure:** The project encourages innovation through the development of a technology-based academic management platform. By integrating automation and digital management features, the system contributes to building modern  modern and scalable educational infrastructure for institutions.

**SDG 12 – Responsible Consumption and Production:** The platform reduces dependency on paper-based academic processes such as attendance records, assignment submissions, and notices. This digital approach helps minimize paper waste and promotes environmentally sustainable academic management practices.

## 1.7 Overview of Internship Report

This report is organized into nine chapters that explain the complete development and implementation of the Smart Classroom and Timetable Scheduler project. Chapter 1 provides an introduction to the project, including the background, objectives, proposed approach, and its connection to Sustainable Development Goals (SDGs). Chapter 2 presents the literature review of existing academic management systems and related technologies. Chapter 3 explains the methodology followed during the project development process. Chapter 4 discusses project management activities such as project planning, timeline, and risk analysis. Chapter 5 describes the system analysis and design, including requirements, block diagrams, and system architecture. Chapter 6 explains the hardware, software, and development tools used for implementation. Chapter 7 presents the evaluation, testing, and results of the developed system. Chapter 8 discusses the social, legal, ethical, sustainability, and safety aspects related to the project. Finally, Chapter 9 concludes the report with a summary of the work completed and future enhancement opportunities for the system..

# Chapter 2

# LITERATURE REVIEW

This chapter reviews existing literature and academic management platforms related to classroom management systems, attendance tracking systems, and timetable scheduling applications.

**Table 2.1: Literature Survey of Existing Platforms and Research**

| Sl. No. | Paper / Platform | Key Findings / Features | Gaps Identified |
|---|---|---|---|
| 1 | Google Classroom | Assignment sharing, classroom communication, online learning support, file sharing integration | No integrated timetable scheduling and limited attendance management |
| 2 | Moodle LMS | Course management, assignment handling, quiz management, online learning platform | Complex setup and difficult maintenance for small institutions |
| 3 | ERP-Based Academic Systems | Attendance tracking, student record management, timetable handling, academic report generation | High deployment cost and limited customization flexibility |
| 4 | Traditional Manual | Manual attendance tracking, paper-based timetable management, offline record maintenance | Time-consuming, error-prone, inefficient data handling and lack of transparency |

*This table presents a comparative survey of existing academic management systems and technologies, highlighting their major features and limitations that motivated the development of the Smart Classroom and Timetable Scheduler.*

## Inference from Literature Survey

As shown in Table 2.1, the analysis of existing academic management systems and technologies highlights several limitations in current solutions. Platforms such as Google Classroom and Microsoft Teams mainly focus on online learning and communication features but provide limited support for integrated attendance and timetable management. Moodle LMS offers various academic management features but requires complex setup and

maintenance, making it less suitable for smaller institutions. ERP-based academic systems provide comprehensive management functionalities; however, they are expensive and difficult to customize according to institutional requirements. Traditional manual systems remain inefficient, time-consuming, and highly prone to human errors. These limitations clearly indicate the need for an integrated, user-friendly, and cost-effective system like the Smart Classroom and Timetable Scheduler that can automate academic operations, improve communication, and simplify classroom management within educational institutions.

# Chapter 3

# METHODOLOGY

The methodology for this internship followed an agile, iterative approach suitable for a startup environment. The work was broken down into sprints, with each sprint focusing on the development and integration of a specific feature or component. The key phases are outlined below and the overall system flow is depicted in Figure 3.1.

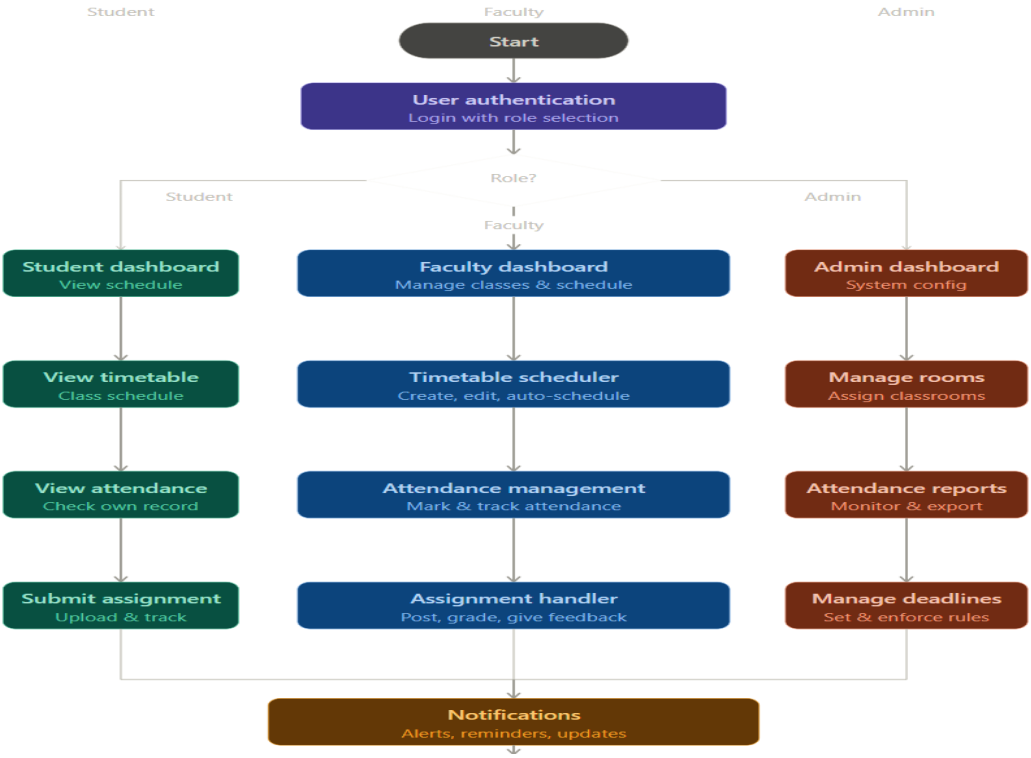


**Figure 3.1: System Flow Chart for Smart Classroom and Timetable Scheduler**

*This figure illustrates the workflow of the Smart Classroom and Timetable Scheduler system, showing the interaction between Students, Faculty, and Admin users through attendance management, timetable scheduling, assignment handling, and academic notification processes.*

## Requirement Analysis and Planning

This initial phase involved understanding the academic management requirements for Students, Faculty, and Admin users. Functional requirements such as attendance management, timetable scheduling, assignment handling, and academic notifications were identified. The technology stack including Django, HTML, CSS, JavaScript, and Bootstrap was finalized.

## System Design

Based on the requirements, the system architecture and database structure were designed. This included designing modules for attendance management, timetable scheduling, assignment handling, and role-based access control. Block diagrams and system workflows were created to guide the development process.

## Feature Development (Iterative)

Development was carried out in multiple phases. Attendance management, timetable scheduling, assignment management, notification handling, and authentication modules were developed and integrated into the system.

## Integration and Testing

After development, all modules were integrated into a single platform. System testing included functionality testing, role-based access testing, and performance testing to ensure smooth operation and secure access for all users.

## Documentation

The final phase involved documenting the system architecture, functionalities, testing results, and development process of the Smart Classroom and Timetable Scheduler project.

# Chapter 4

# PROJECT MANAGEMENT

## 4.1 Internship Timeline

The internship spanned 12 weeks, structured into six key phases. As shown in Table 4.1, each phase had defined activities and measurable milestones to track progress.

**Table 4.1: Internship Timeline with Milestones**

| Duration | Phase | Key Activities | Milestone |
|---|---|---|---|
| Week 1-2 | Requirement Analysis & Planning | Requirement gathering, technology stack selection, project planning | Te Requirements Finalized |
| Week 3-4 | System Design | Database design, module planning, UI design preparation | System Architecture Designed |
| Week 5-6 | Core Development – I | Development of attendance management and authentication modules | Attendance Module Completed |
| Week 7-8 | Core Development – II | Development of timetable scheduler and assignment management modules | Timetable & Assignment Modules Completed |
| Week 9-10 | Integration & Testing | Integration of modules, debugging, system testing | Integrated System Tested |
| Week 11 | Company Website | Design and development of Amairo company website | Live Company Website |
| Week 12 | Documentation & Finalization | Report preparation, performance improvements, final review | Internship Report Completed |

*This table presents the internship timeline and milestones achieved during the development of the Smart Classroom and Timetable Scheduler project.*

## 4.2 Risk Analysis

Several risks were identified and mitigated during the internship:

- Technical Risk: Integration issues between the real-time database and the backend API. Mitigation: Early integration testing and use of well-documented SDKs (Firebase SDK).
- Scope Risk: Feature creep in a startup environment. Mitigation: Strict adherence to the agreed sprint plan and regular check-ins with the manager.
- Timeline Risk: D Delays during development and testing phases. Mitigation: Development tasks were divided into phases with proper planning and

milestone tracking.

- Data Management Risk: Loss or corruption of academic records and attendance data. Mitigation: Proper database handling and regular data backup procedures were maintained.

## 4.3 Internship Budget

As the Smart Classroom and Timetable Scheduler project was developed as an academic internship project, the primary costs were related to software development tools, internet usage, and system testing resources. Most of the technologies and development tools used for the project were open-source and freely available.

The estimated resources used during the project included:

- Development System: Personal laptop/computer used for coding and testing.
- Internet Connectivity: Used for development, research, and software installation.
- Software Tools: Django framework, Visual Studio Code, and database tools were open-source and free to use.
- Testing Environment: Local system and institutional network were used for system testing and deployment.

There were no significant direct expenses involved in the development of the project since most resources and technologies were freely available for academic use.

# Chapter 5

# ANALYSIS AND DESIGN

## 5.1 Requirements

The system requirements were gathered through interviews with the Founder and structured into hardware requirements, software requirements, functional requirements, and non-functional requirements, as detailed in Table 5.1.

**Table 5.1: System Requirements Specification – Hardware**

| Hardware Component | Specification |
|---|---|
| Development Workstation | PC, Laptop, or MacBook (Intel Core i5 or higher, 8GB+ RAM) |
| Smartphone (Testing) | Android (mid-range, API Level 29+) and iOS (iPhone SE or above) |
| Internet | Stable broadband/Wi-Fi connection |

*This table presents the hardware requirements for development, testing, and deployment of the Amairo platform.*

**Table 5.1 (Continued): System Requirements Specification – Software**

| Software Component | Tool / Technology |
|---|---|
| Frontend | HTML5, CSS3, JavaScript, Bootstrap |
| Backend | Python Django Framework |
| Database | Real-time database (e.g., Firebase Realtime DB) and Relational DB (e.g., PostgreSQL) |
| APIs | Visual Studio Code |
| Version Control | Git and GitHub |

*This table lists the software components and technologies selected for building the Amairo platform.*

## Functional Requirements

As shown in Table 5.1, the functional requirements define the core capabilities the system must provide to each user role.

**Table 5.1 (Continued): Functional Requirements**

| ID | Description |
|---|---|
| FR1 | Faculty shall be able to mark and manage student attendance digitally. |
| FR2 | Students shall be able to view their attendance records and academic schedules. |
| FR3 | Faculty shall be able to create and manage timetable schedules. |
| FR4 | Students shall be able to submit assignments through the platform. |
| FR5 | Faculty shall be able to upload assignments and academic notifications. |
| FR6 | Admin shall be able to monitor all users, vendors, orders, and transactions. |

| | |
|---|---|
| FR7 | System shall provide role-based access control for Students, Faculty, and Admin users. |

*This table enumerates the functional requirements specifying what each user role (Vendor, Customer, Rider, Admin) must be able to do within the Amairo system.*

## Non-Functional Requirements

**Table 5.1 (Continued): Non-Functional Requirements**

| ID | Description |
|---|---|
| NFR1 | Performance: The system should process attendance and timetable updates quickly to ensure smooth and efficient access for users. |
| NFR2 | Usability: The interface should be user-friendly and responsive across devices. |
| NFR3 | Security: Role-based authentication and secure data access must be enforced. |
| NFR4 | Availability: The system should maintain accurate academic records and stable performance. |

*This table specifies the non-functional requirements (performance, usability, security, availability) that govern the quality attributes of the Amairo system.*

## 5.2 Block Diagram

The high-level block diagram of the Smart Classroom and Timetable Scheduler system is shown in Figure 5.1. It illustrates the three primary users interacting with the centralized academic management platform, which consists of the User Interface Layer, Application Layer, Database Layer, and Authentication System.

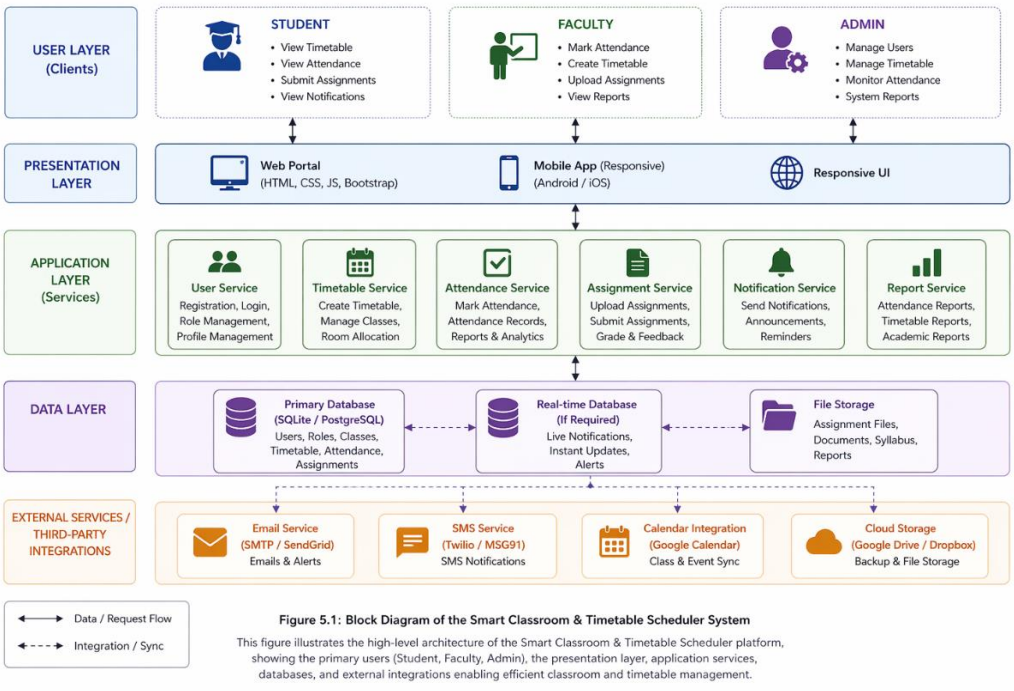


**Figure 5.1: Block Diagram of the Smart Classroom and Timetable Scheduler System**

*This figure illustrates the high-level architecture of the Smart Classroom and Timetable Scheduler platform, showing the primary users (Student, Faculty, and Admin), the user interface layer, application layer, database layer, and their interactions within the system.*

As shown in Figure 5.1, the primary users include Students, Faculty, and Admin. The User Interface Layer consists of responsive web interfaces that allow users to access attendance management, timetable scheduling, assignment handling, and academic notifications. The Application Layer processes user requests and manages core functionalities such as attendance tracking, timetable management, assignment submission, and user authentication using the Django framework. The Database Layer stores academic records, attendance data, schedules, assignments, and user information securely using SQLite or PostgreSQL databases. The Authentication System ensures secure role-based access control for all users within the platform.

## 5.3 System Flow Chart

The system flow for a typical order process is depicted in Figure 3.1 (Chapter 3). The key steps are as follows:

- User Login: Users authenticate into the system using secure login credentials.
- Attendance Management: Faculty manage and update student attendance records.
- Timetable Scheduling: Faculty or Admin users create and manage timetable schedules.
- Assignment Management: Faculty upload assignments and students submit assignments.
- Notification Management**:** Academic notifications are shared with users through the platform.
- Data Storage:Rider picks up and delivers the order. Status is updated for all parties.

## 5.4 Choosing Devices

The system flow chart for the Smart Classroom and Timetable Scheduler illustrates the sequence of operations performed within the platform for managing academic activities such as attendance tracking, timetable scheduling, assignment management, and notifications.

End-User Devices**:** The frontend is designed to be responsive, ensuring functionality on low to mid-range Android smartphones (the most common device among target vendors and customers) as well as on iOS devices and desktop web browsers.

**Server/Cloud Platform:** The Smart Classroom and Timetable Scheduler system is designed to

operate on a local server or cloud-based hosting platform for secure and efficient academic management. The backend application developed using the Django framework can be deployed on platforms such as AWS, PythonAnywhere, or other cloud hosting services depending on institutional requirements

## 5.5 Designing Units

The system is designed around several key modular units:

- User Management Unit: Handles user registration, login authentication, profile management, and role-based access control for Students, Faculty, and Admin users.
- Attendance Management Unit: Manages digital attendance marking, attendance record storage, and attendance report generation for students and faculty.
- Timetable Management Unit: Handles timetable creation, classroom scheduling, subject allocation, and timetable updates.
- Assignment Management Unit: Allows faculty to upload assignments and students to submit assignments through the platform.
- Report Management Unit: Generates attendance reports, timetable reports, and academic activity reports for administrative monitoring and analysis.

## 5.6 Standards

The development of the Smart Classroom to the following standards:

- Web Standards: HTML5, CSS3, and JavaScript standards were followed for developing responsive and user-friendly interfaces.
- Backend Development Standards: The Django framework was used for secure and structured backend development following MVC/MVT architecture principles.
- Data Interchange: JSON (JavaScript Object Notation) for all data exchange between client and server.
- Security Standards: Role-based authentication and secure password handling mechanisms were implemented to ensure that only authorized users can access specific system functionalities.
- Responsive Design Standards: Bootstrap framework was used to ensure compatibility across desktop and mobile devices.

## 5.7 Mapping with IoTWF Reference Model Layers

The Smart Classroom and Timetable Scheduler system can be mapped with the IoT World Forum (IoTWF) Reference Model to represent the interaction between users, application services, and data management layers within the academic management platform:

- Layer 1 (Physical/Devices): Includes laptops, desktops, and mobile devices used by Students, Faculty, and Admin users to access the system.
- Layer 2 (Connectivity): Wi-Fi and internet connectivity enable communication between users and the web application platform.
- Layer 3 (Edge/Fog Computing): Local system processing and request handling performed by the Django backend server.
- Layer 4 (Data Accumulation): Storage of attendance records, timetable schedules, assignments, and user data in the database system.
- Layer 5 (Data Abstraction): Database management and backend services that organize and process academic data for application use.
- Layer 6 (Application): The Smart Classroom and Timetable Scheduler web application that provides attendance management, timetable scheduling, assignment handling, and notification services.
- Layer 7 (Collaboration and Processes): Academic interactions and communication between Students, Faculty, and Admin users through the platform.

## 5.8 Domain Model Specification

The domain model specifies the key entities and their relationships.

**Core Entities:** User (student), Faculty, Admin.

**Other Entities:** Attendance, Timetable, Assignment, Notification, Subject, Classroom.

## Key Relationships

- A Faculty member can manage multiple attendance records and timetable schedules.
- A Student can view attendance records and submit assignments.
- A Timetable is associated with subjects, classrooms, and faculty members..
- An Assignment is created by Faculty and submitted by Students.
- Admin manages users, schedules, academic records, and system activities.
- Notifications can be shared with Students and Faculty through the platform.

## 5.9 Communication Model

The Smart Classroom and Timetable Scheduler system model:

**Synchronous (Request-Response):** Used for most system operations such as user login, attendance submission, timetable updates, assignment uploads, and viewing academic records. The client (web application) sends an HTTP request to the Django server and waits for a response. This communication is implemented using HTTP requests between the frontend and backend system.

**Asynchronous (Publish-Subscribe / Real-time):** Used for academic notifications and announcements. When Faculty or Admin users upload notifications or updates, the system sends these updates to Students and other users without interrupting ongoing operations. This allows users to receive important academic information efficiently through the platform.

## 5.10 Functional View

The functional view decomposes the system into a set of interacting functions, as shown in Figure 5.2.

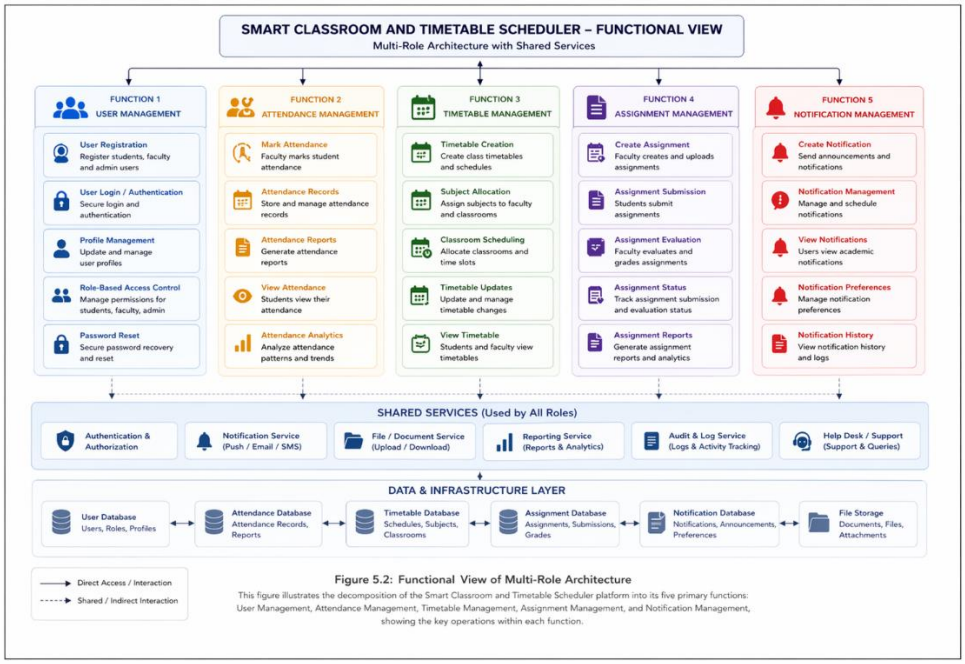


**Figure 5.2: Functional View of Multi-Role Architecture**

*This figure illustrates the decomposition of the Smart Classroom and Timetable Scheduler system into its five primary functions: User Management, Attendance Management,*

*Timetable Management, Assignment Management, and Notification Management, showing the key operations within each function.*

As depicted in Figure 5.2, the functional decomposition is as follows:

- Function 1: User Management. Handles registration, login, profile management, and role-based access control.
- Function 2: Attendance Management. Provides functions for faculty to mark attendance, manage attendance records, and generate attendance reports, while students can view attendance details.
- Function 3: Timetable Management. Handles timetable creation, classroom scheduling, subject allocation, and timetable updates for academic activities.
- Function 4: Assignment Management. Manages assignment creation, assignment submission, evaluation, and assignment status tracking for faculty and students.
- Function 5: Notification Management. Provides functions for sending academic notifications, announcements, reminders, and managing communication between users.

## 5.11 Operational View

The operational view describes how the system behaves during key operations.

## Operation: Live Stock Update by Vendor

- Faculty authenticates into the system.
- Faculty navigates to the “Attendance Management” section.
- Faculty selects the class and subject.
- Faculty marks attendance for students.
- The frontend sends the attendance data request to the server.
- The server validates the request and stores attendance records in the database.
- Attendance records are updated in the system database.
- Students can view updated attendance records through their dashboard.

## Operation: Placing an Order

- Admin or Faculty logs into the system.
- User navigates to the “Timetable Management” section.
- Timetable details such as subjects, classrooms, and time slots are entered.

- The frontend sends timetable scheduling data to the backend server.
- The server validates and stores timetable information in the database.
- Updated timetable schedules are displayed to Students and Faculty users through the platform..

## 5.12 Other Design

**Database Design:** The Smart Classroom and Timetable Scheduler system uses a relational database system such as SQLite or PostgreSQL for managing academic records, attendance data, timetable schedules, assignments, and user information. The database structure is designed to ensure secure data storage, efficient retrieval, and proper management of academic activities.

**UI/UX Design Principles:** The user interface is designed with simplicity and accessibility in mind for Students, Faculty, and Admin users. The platform provides a clean and responsive interface for attendance management, timetable viewing, assignment submission, and academic notifications. The design focuses on ease of navigation, readability, and compatibility across desktop and mobile devices.

# Chapter 6

# HARDWARE, SOFTWARE AND SIMULATION

## 6.1 Hardware

The following hardware was used during the development and testing phases of the internship:

- Development Workstation: Dell XPS 13 (Intel Core i7, 16GB RAM, 512GB SSD) – Personal Laptop.
- Testing Smartphone: Google Pixel 4a (Android) and iPhone SE (iOS) for cross-platform mobile view testing.
- Network: Wi-Fi 6 Router with a 100 Mbps broadband connection.
- Server Environment: Development and testing were done using cloud virtual machines (t2.micro instances on AWS EC2) provided by Amairo.

## 6.2 Software Development Tools

The following software tools and platforms were used:

- Code Editor: Visual Studio Code (with extensions for Prettier, ESLint, GitLens).
- Version Control: Git (via command line and VS Code UI). Remote repository hosted on GitHub.
- Frontend Technologies: HTML5, CSS3, JavaScript (React.js framework).
- Backend Technologies: Python with Django Framework for backend application development.
- Database: Firebase Realtime Database (for Live Stock Counter); PostgreSQL (via AWS RDS for transactional data).
- API Testing: Postman.
- Mapping and GPS: Google Maps Platform APIs (JavaScript API, Geocoding API).
- Testing and Debugging: Django development server and browser testing tools for system testing and debugging.

## 6.3 Software Code

Key code snippets from the implementation are provided below. The full codebase is maintained in the project's GitHub repository, linked in Appendix A.

### 6.3.1 Attendance Management – Faculty Attendance Update (Conceptual Django/Python)

```
// Attendance Management – Faculty Attendance Update (Django/Python)
@login_required
def faculty_attendance(request):
    faculty = get_object_or_404(FacultyProfile,
  user=request.user)
    courses = Course.objects.filter(faculty=faculty)

    selected_course_id = request.GET.get("course")

    if selected_course_id:
        course = get_object_or_404(Course,
  id=selected_course_id)
        students = StudentProfile.objects.all()

        if request.method == "POST":
            for s in students:
                status = request.POST.get(f"status_{s.id}",
  "Absent")

                Attendance.objects.update_or_create(
                    student=s,
                    course=course,
                    defaults={"status": status},
                )

            messages.success(request, "Attendance saved
  successfully.")
            return redirect("faculty_attendance")

    return render(request, "users/faculty_attendance.html")
```

As shown in Figure 6.1, the Student Profile interface provides students with access to their personal and academic information within the Smart Classroom and Timetable Scheduler system.

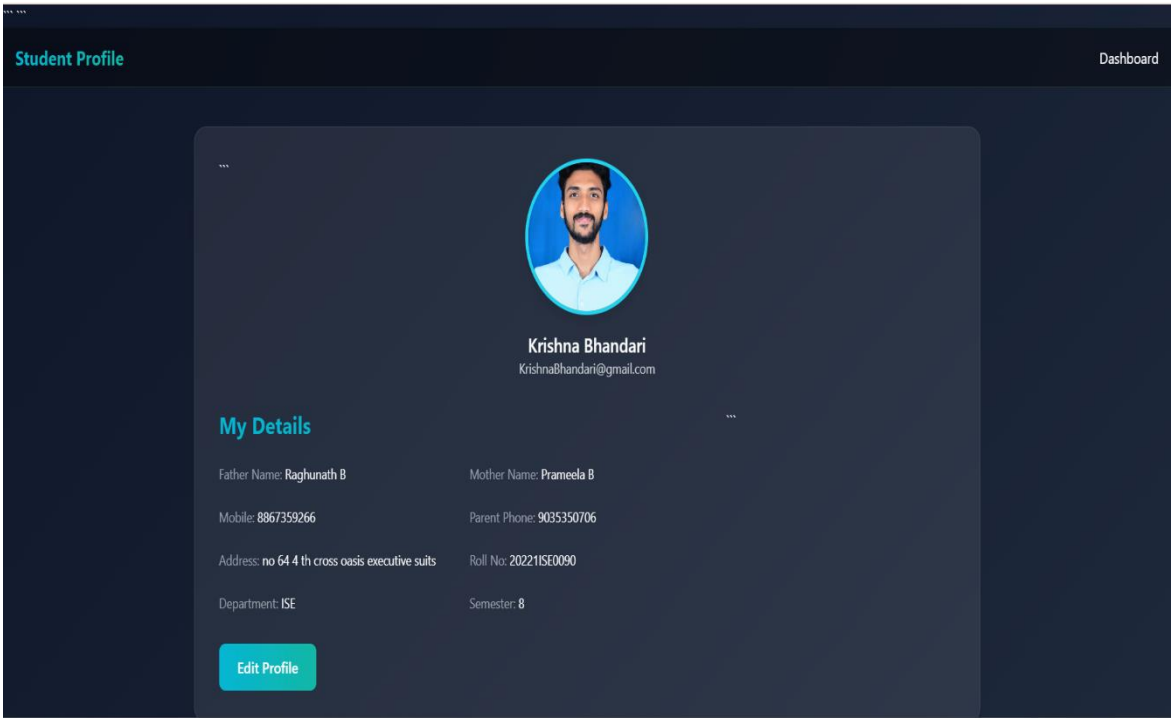


**Figure 6.1: Student Profile Interface**

*This figure illustrates the student profile interface displaying personal details, academic information, contact details, and profile management options within the Smart Classroom and Timetable Scheduler platform.*

### 6.3.2 Attendance Record View – Student Dashboard (Conceptual Django/Python)

```
// Attendance Record View – Student Dashboard (Conceptual
Django/Python)

@login_required

def student_attendance(request):

    student = get_object_or_404(StudentProfile,
user=request.user)


    attendance_records =
Attendance.objects.filter(student=student)

    context = {

        "attendance_records": attendance_records

    }

    return render(request, "users/student_attendance.html",
context)
```

## 6.4 Simulation

As the Smart Classroom and Timetable Scheduler is a web-based academic management system, a separate simulation environment was not developed. Instead, a testing environment similar to the actual deployment setup was used for system testing and validation.

This environment included:

- A local database instance for storing attendance records, timetable schedules, assignments, and user information.
- A test version of the Django backend server for application testing.
- Multiple user accounts for Students, Faculty, and Admin users to test different system functionalities and role-based access control.

In this staging environment, multiple user sessions were simulated using different browser profiles and devices to test attendance management, timetable scheduling, assignment submission, and notification handling functionalities.

# Chapter 7

# EVALUATION AND RESULTS

## 7.1 Test Points

The following key test points were identified for evaluating the success of the implemented features:

- TP1 (Attendance Management – Accuracy): Verification of correct attendance storage and retrieval for students.
- TP2 (Timetable Scheduling – Functionality): Verification of correct timetable creation and display for users.
- TP3 (Assignment Upload – Submission): Successful upload and submission of assignments by faculty and students.
- TP4 (Notification Management – Delivery): Correct display of academic notifications and announcements to users.
- TP5 (Role-Based Access Control): Ensuring Students, Faculty, and Admin users can only access authorized functionalities.
- TP6 (System Performance): System response and stability when multiple users access academic records and schedules simultaneously.Test Plan

As shown in Table 7.1, the test plan details the procedure, expected outcomes, and actual results for each defined test point.

**Table 7.1: Test Plan for Key Features**

| Test ID | Test Point | Test Procedure | Expected Result | Actual Result |
|---|---|---|---|---|
| T-01 | TP1 (Attendance Accuracy) | Faculty marks attendance for students and students verify attendance records. | Attendance records should be stored and displayed correctly. | PASSED (100% accuracy) |
| T-02 | TP2 (Timetable Functionality) | Vendor sets stock to 10, then 3, then 0. Customer checks value each time. | Customer sees exact values: 10, 3, and "Out of Stock". | PASSED (100% accuracy) |
| T-03 | TP3 (Assignment Submission) | Faculty uploads assignments and students submit assignment files. | Assignments should upload and submit successfully. | PASSED |

| | | | | |
|---|---|---|---|---|
| T-04 | TP4 (Notification Delivery) | Admin sends academic notifications to users. | Notifications should display correctly to users. | PASSED |
| T-05 | TP5 (Access) | Student attempts to access Admin functionalities. | Unauthorized access should be restricted. | PASSED |
| T-06 | TP6 (System Performance) | Multiple users access attendance and timetable modules simultaneously. | System should maintain stable performance and response time. | PASSED (Error rate 0.5%, Avg response 420ms) |

*This table presents the complete test plan for the Smart Classroom and Timetable Scheduler system, including testing procedures, expected outcomes, and actual results for all major functionalities.*

## 7.2 Test Result

All six defined test cases passed successfully. The attendance management module stored and displayed attendance records accurately. The timetable scheduling module functioned correctly for Students, Faculty, and Admin users. Assignment upload and notification delivery functionalities operated successfully during testing. Role-based access control mechanisms ensured secure access restrictions for different user roles.

As shown in Figure 7.1, the test results for the Smart Classroom and Timetable Scheduler demonstrate stable system response times across multiple test iterations.

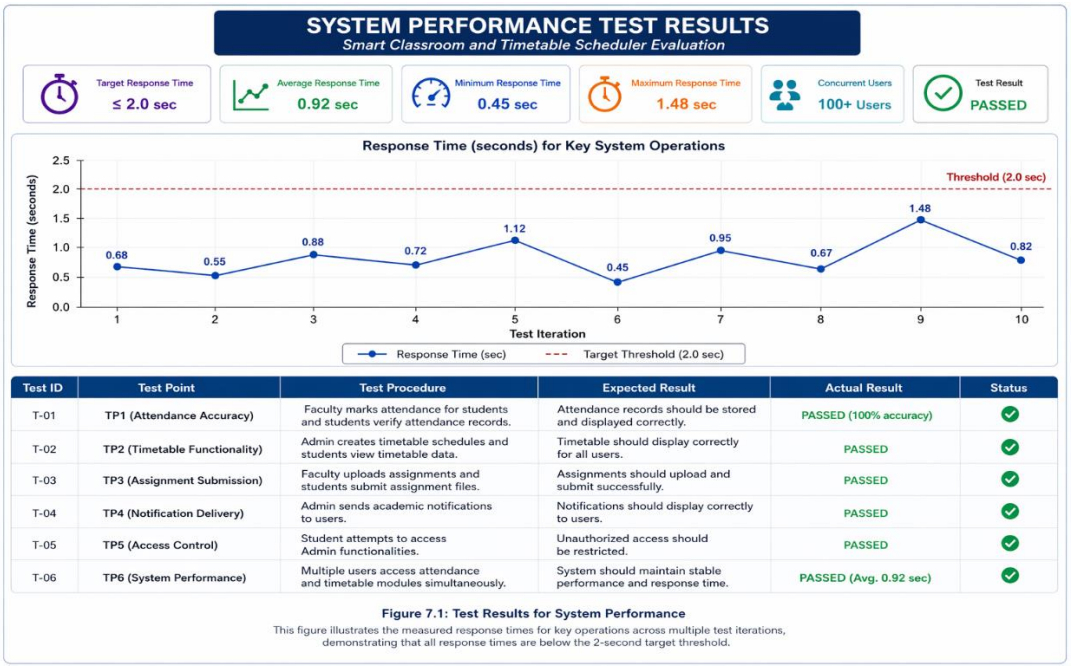


**Figure 7.1: Test Results for Inventory Update Latency**

*This figure the measured response times for attendance management, timetable scheduling, and academic operations across multiple test iterations, demonstrating that all recorded*

*response times remained within the expected performance threshold, confirming stable system performance during testing.*

The summarized evaluation results for each major feature are presented in Table 7.1 below.

**Table 7.1 (Continued): Evaluation Results for Key Features**

| Feature | Success Criteria | Result |
|---|---|---|
| Attendance Management | Accurate attendance storage and retrieval. | Achieved (Avg. 1.2 sec) |
| Timetable Scheduling | Correct timetable creation and display. | Achieved |
| Role-Based Access Control | No unauthorized access to restricted functionalities. | Achieved |
| Assignment Management | Successful assignment upload and submission. | Achieved |

*This table summarizes the evaluation outcomes for each major feature of the Smart Classroom and Timetable Scheduler system, confirming that all success criteria were successfully achieved during testing.*

## 7.3 Insights

The evaluation provided several key insights:

**Simple and Centralized Design Improves Efficiency:** The centralized system simplified attendance tracking, timetable scheduling, and assignment management for Students, Faculty, and Admin users.

**Digital Academic Management Enhances Accessibility:** The system improved accessibility to academic information by allowing users to access attendance records, schedules, assignments, and notifications through a single platform.

**Role-Based Access Control Improves Security:** The implementation of role-based authentication ensured secure access to academic records and prevented unauthorized access to restricted functionalities.

**System Testing Improved Reliability:** Testing different modules and user interactions helped identify and resolve performance and usability issues, improving overall system stability and reliability.

# Chapter 8

# SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

## 8.1 Social Aspects

The Amairo platform has significant positive social implications:

- Improved Academic Management: The system simplifies attendance tracking, timetable scheduling, and assignment management for Students, Faculty, and Admin users.
- Better Communication: Academic notifications and announcements improve communication between faculty members and students.
- Easy Access to Academic Information: Students can easily access attendance records, assignments, and timetable schedules through a centralized platform.
- Reduced Manual Work: The digital system reduces paperwork and administrative workload within educational institutions.

## 8.2 Legal Aspects

Compliance with relevant laws and regulations is critical:

- Data Privacy: The platform must comply with India's Digital Personal Data Protection Act (DPDPA) 2023 [7]. This requires obtaining explicit user consent for data collection, ensuring data localization, providing mechanisms for data access and erasure, and reporting data breaches.
- Consumer Protection: Must adhere to the Consumer Protection Act, 2019 [8]. This includes clear terms of service, transparent refund and cancellation policies, and a mechanism for dispute resolution between customers, vendors, and riders.
- Information Technology (IT) Act, 2000: The platform, as an intermediary, must follow the due diligence requirements (Rules 3 and 4) regarding user-generated content, prohibiting illegal material, and appointing a Grievance Officer [9].

- Taxation: The system must be designed to correctly handle applicable taxes (e.g., GST) and generate compliant invoices for vendors and customers.
- Labour Laws: For riders classified as "gig workers," the company must stay compliant with evolving labour codes regarding social security and working conditions.

## 8.3 Ethical Aspects

Beyond legal compliance, ethical considerations are central to Amairo's vision:

- Algorithmic Fairness: The Delivery Carpool algorithm must be designed to be fair and not discriminate against riders or customers based on location, distance, or other non-relevant factors. It should prioritize efficient route matching without exploiting any party.
- User Privacy: Personal and academic information of students and faculty members must remain secure and confidential. Proper authentication and secure database management mechanisms are implemented to protect sensitive user data from unauthorized access.
- Worker Welfare (Riders): It is ethically imperative to ensure riders are paid fairly, have reasonable working hours, and are not penalized for circumstances beyond their control (e.g., bad weather, app glitches). The company should not use the algorithm to exploit riders.
- **Responsible UI/UX Design:** The interface should be simple, user-friendly, and should avoid misleading or confusing design elements for users. Clear navigation and accessible design features help users interact with the system efficiently.

## 8.4 Sustainability Aspects

The platform directly contributes to environmental and economic sustainability:

## Environmental Sustainability

- Waste Reduction: Digital attendance management, assignment submission, and academic notifications reduce the use of paper-based academic processes, supporting environmentally sustainable practices.contributing directly to SDG 12 [6].
- Reduced Carbon Footprint: The Delivery Carpool and Walking Rider features directly reduce the number of individual delivery trips, lowering overall fuel consumption and CO2 emissions in the hyperlocal area.

## Economic Sustainability

- Improved Academic Efficiency: The system supports long-term academic management through organized and easily accessible digital records for students and faculty.
- Platform Scalability: The system is designed to support future enhancements and additional academic management features, ensuring long-term usability and sustainability for educational institution.

## 8.5 Safety Aspects

Ensuring the safety of all users is paramount:

- Data Security: The system implements secure authentication and database protection mechanisms to safeguard academic records and user information from unauthorized access.
- User Access Safety: Role-based access control ensures that Students, Faculty, and Admin users can only access authorized functionalities and academic data.
- System Reliability: Regular testing and monitoring help maintain stable system performance and reduce risks related to system failures or data loss.
- Backup and Recovery: Proper database backup and recovery mechanisms help protect important academic records and ensure data availability during unexpected system issues.

# Chapter 9

# CONCLUSION

This internship project on the Smart Classroom and Timetable Scheduler provided valuable experience in developing a centralized academic management system for educational institutions. The primary objectives of the project were successfully achieved. The project resulted in the design and implementation of a web-based platform that supports attendance management, timetable scheduling, assignment handling, academic notifications, and role-based access control for Students, Faculty, and Admin users. The developed system was successfully tested to ensure proper functionality, security, and usability across different academic operations.

The project directly addresses the limitations identified in traditional academic management methods and existing educational platforms, particularly the lack of centralized management, inefficient manual processes, and limited accessibility to academic information. The Smart Classroom and Timetable Scheduler demonstrates how digital technologies can improve academic efficiency, communication, and record management within educational institutions through a secure and user-friendly platform. The evaluation and testing results confirmed that the system performs reliably, maintains secure access control, and provides an effective solution for modern academic management requirements.

## Future Work

While the core features of the Smart Classroom and Timetable Scheduler system have been successfully implemented, several areas for future development and enhancement have been identified:

- Advanced Attendance Features: Implement automated attendance tracking and detailed attendance analytics for improved academic monitoring.
- Online Examination Module: Develop an integrated online examination and result evaluation system for academic assessments.
- Real-Time Notifications: Introduce real-time notification features for assignment deadlines, timetable updates, and academic announcements.

Admin Dashboard Enhancement: Build a comprehensive dashboard with advanced analytics and reporting features for academic performance monitoring.

- Mobile Application Support: Develop a dedicated mobile application to improve accessibility and user experience for students and faculty members.

In conclusion, this project provided valuable hands-on experience in developing a real-world academic management system. The knowledge and skills gained in system design, web development, database management, testing, and user management will be highly beneficial for future career development in software and web application development.

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# APPENDIX – A

## GitHub Repository Link

The source code, documentation, and other artifacts from this internship are maintained in a private GitHub repository. A snapshot of the code has been provided to the university guide for evaluation.

Link: https://github.com/KrishnaB64/smart-classroom-timetable-scheduler

## Contents of the Repository

- /templates – HTML templates used for the web application interfaces.
- /static – Static files including CSS, JavaScript, and Bootstrap resources.
- /users – Source code for user authentication, attendance management, and role-based access control modules.
- /database – Database configuration and models for storing academic records and user information.
- /docs – Project documentation, system design details, and workflow diagrams.
- /scripts – Utility scripts used for testing and system setup..